10/09/25

Operations and
Research 19MNG338

Optimizing Citi-Bike Rebalancing with a Routing and cost effective Model

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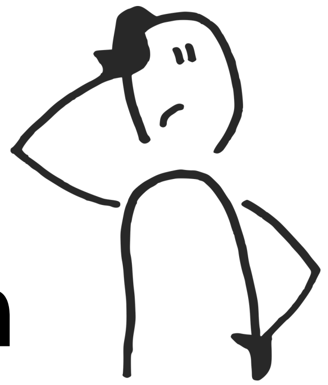
Overview

01 Introduction

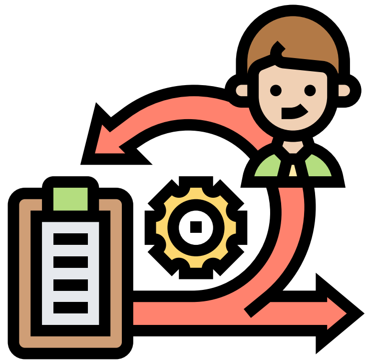


→ Motivation behind the work

02 Dataset

→ Insights from **citi bike** dataset

03 Methodology

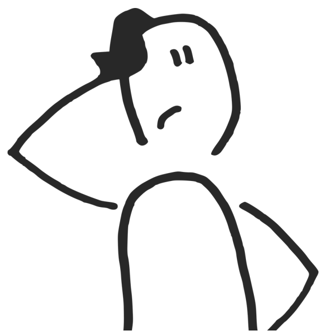


→Implementing algorithms

01
Introduction

Why is this problem important and who is affected by this

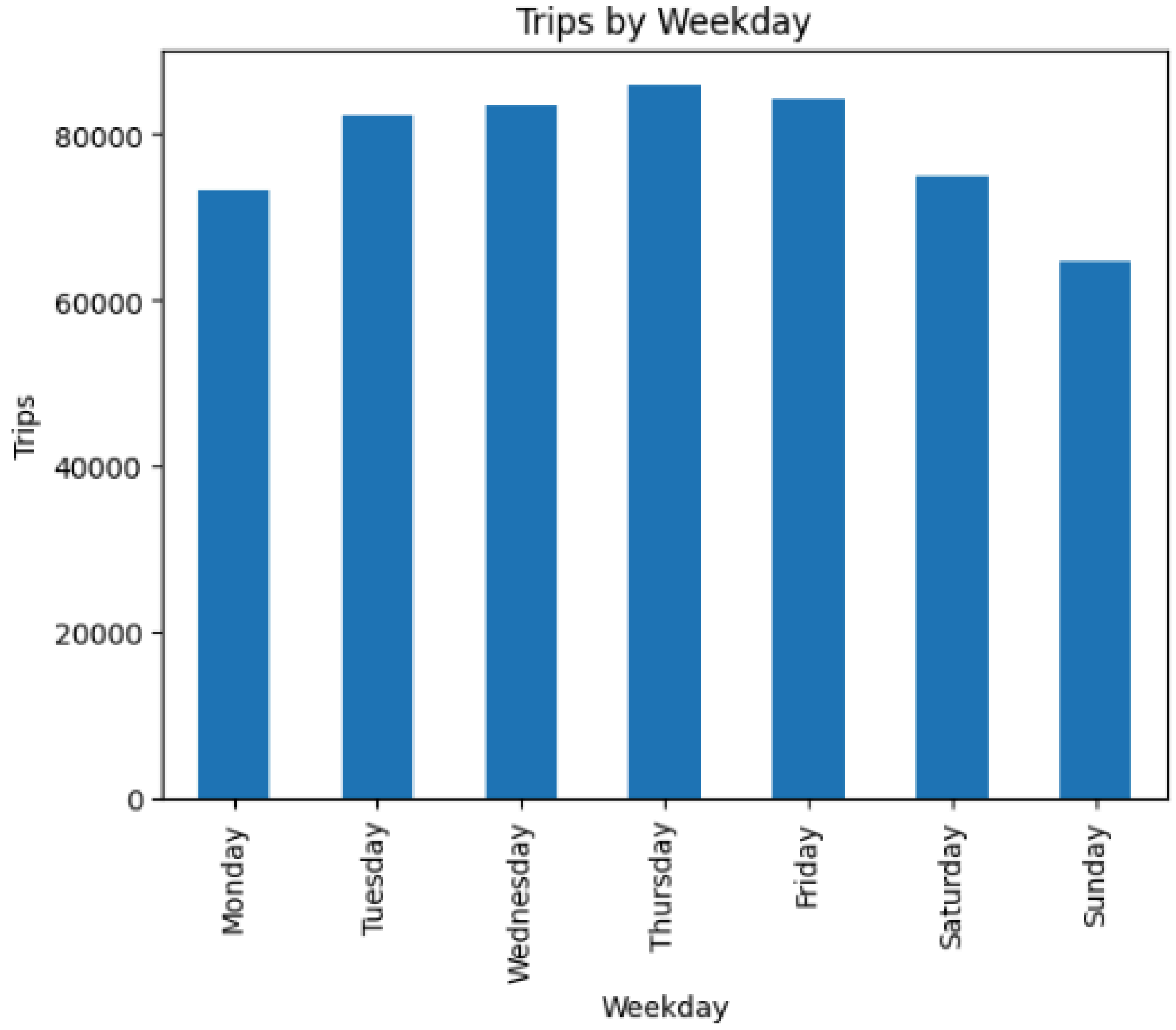
Introduction



Ensuring Bike Availability and Operational Efficiency

Bike-sharing is a critical part of modern urban transit, offering a flexible and green transportation option.

Asymmetric user patterns (e.g., morning commutes downhill or to transit hubs) create severe imbalances. Some stations become empty ("drains"), while others become completely full ("pools"), leading to user frustration and lost revenue.



Problem Statement

Determine the optimal number of bikes to transport from station j to station i such that the demand at station i is fully satisfied, while selecting the appropriate vehicles to minimize total transportation cost.

Decision Variables

x_i: Bikes allocated to station i (integer, ≥0)

$\Delta_i = x_i - s_i$: Net bikes to deliver to station i (s_i = starting inventory at station i)

z_{ijk}: Number of bikes transported from station i to station j by truck k

y_{ijk}: Binary variable; 1 if truck k goes from i to j, else 0

l_{ik}: Load on truck k after visiting station i

t_{ik}: Arrival time of truck k at station i

K: Number of available trucks

C_k: Capacity of truck k (number of bikes)

Objective Function

$$\text{Minimize } \sum_{ijk} \text{Cost}_{ijk} \times z_{ijk} + \sum_{i} \text{Penalty}_i \times (\text{Deficit}_i)$$

Primary Constraints

Rebalancing is only allowed between 1am-5am, no moves at other hours.

No move between two stations can exceed the selected vehicle's capacity.

No station can receive more bikes than it has docks.

No station can have zero or less bikes at any given point

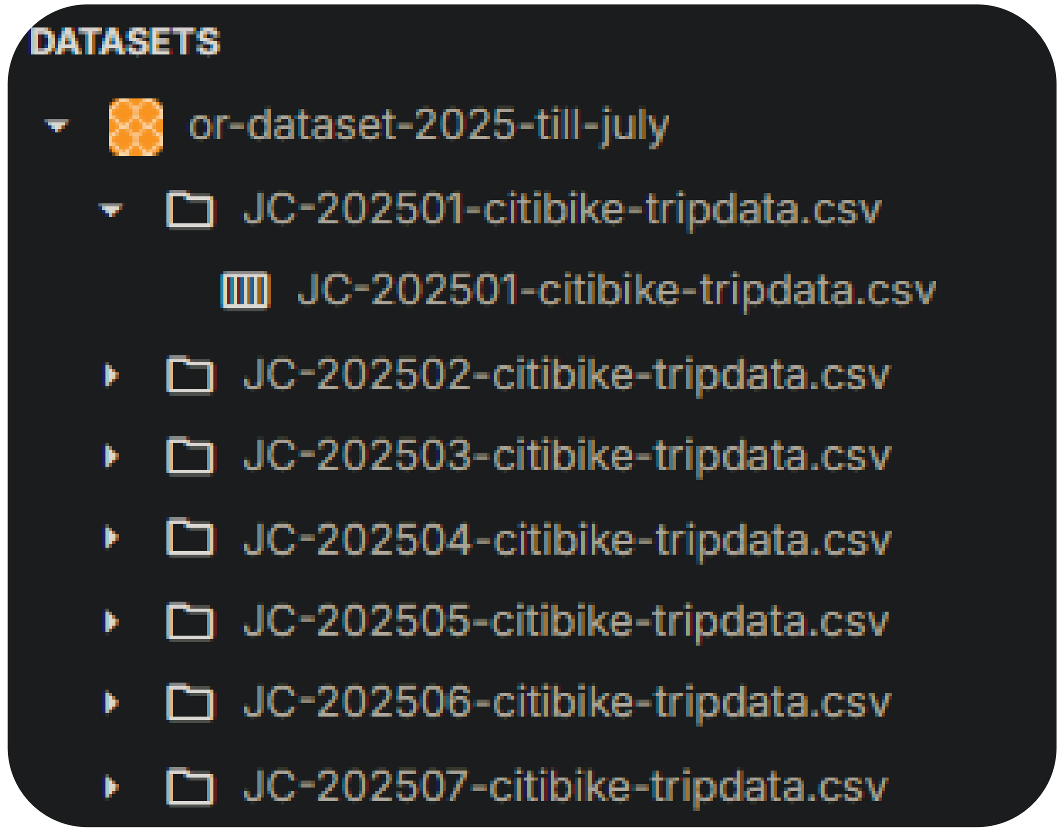
Multiple Vehicle Types Allowed per Move

Fairness/Priority Penalty for Shortages at Remote Stations

02 Dataset

EDA

EDA



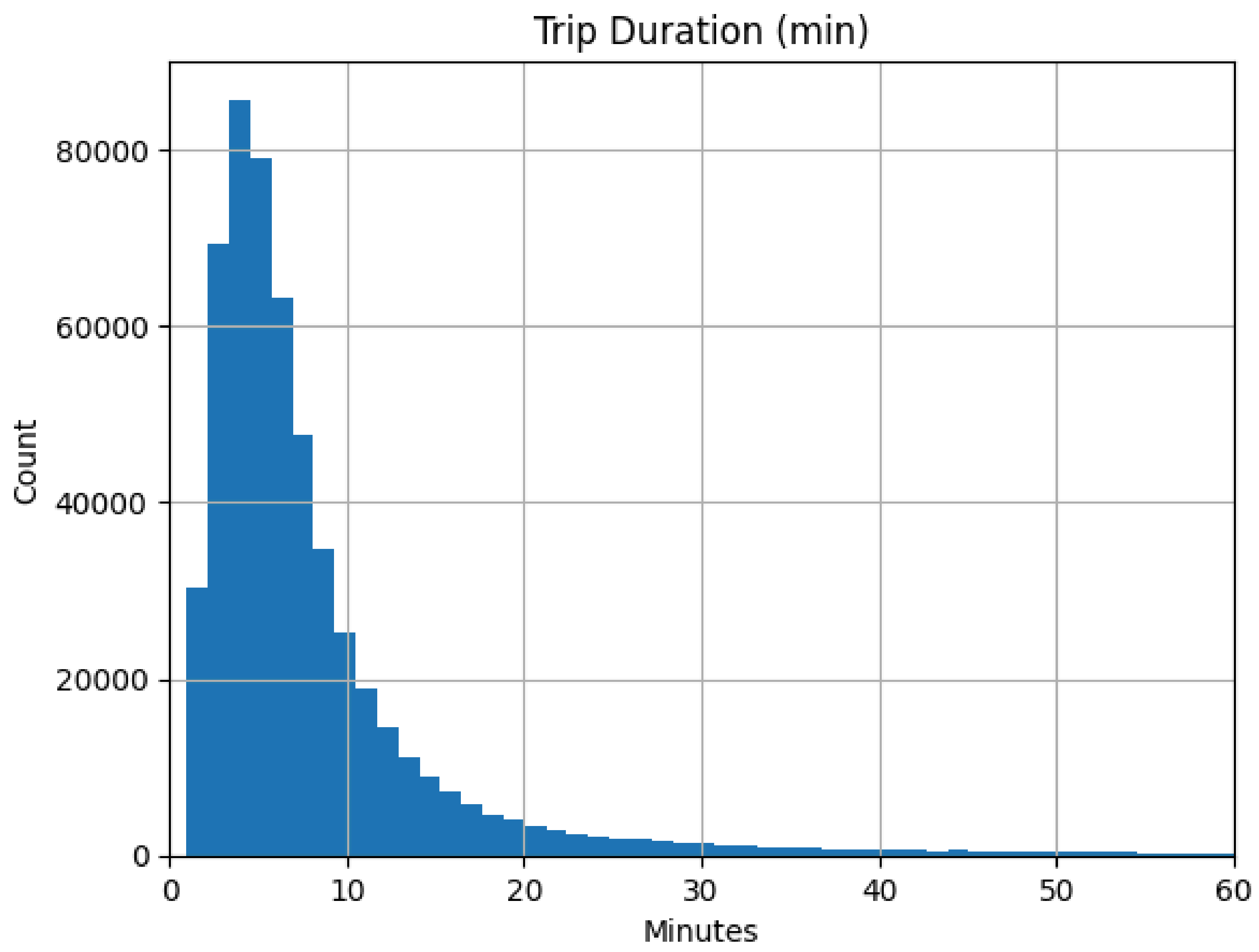
7 months of data

data_shape = (rows = 5,48,789,cols = 13)

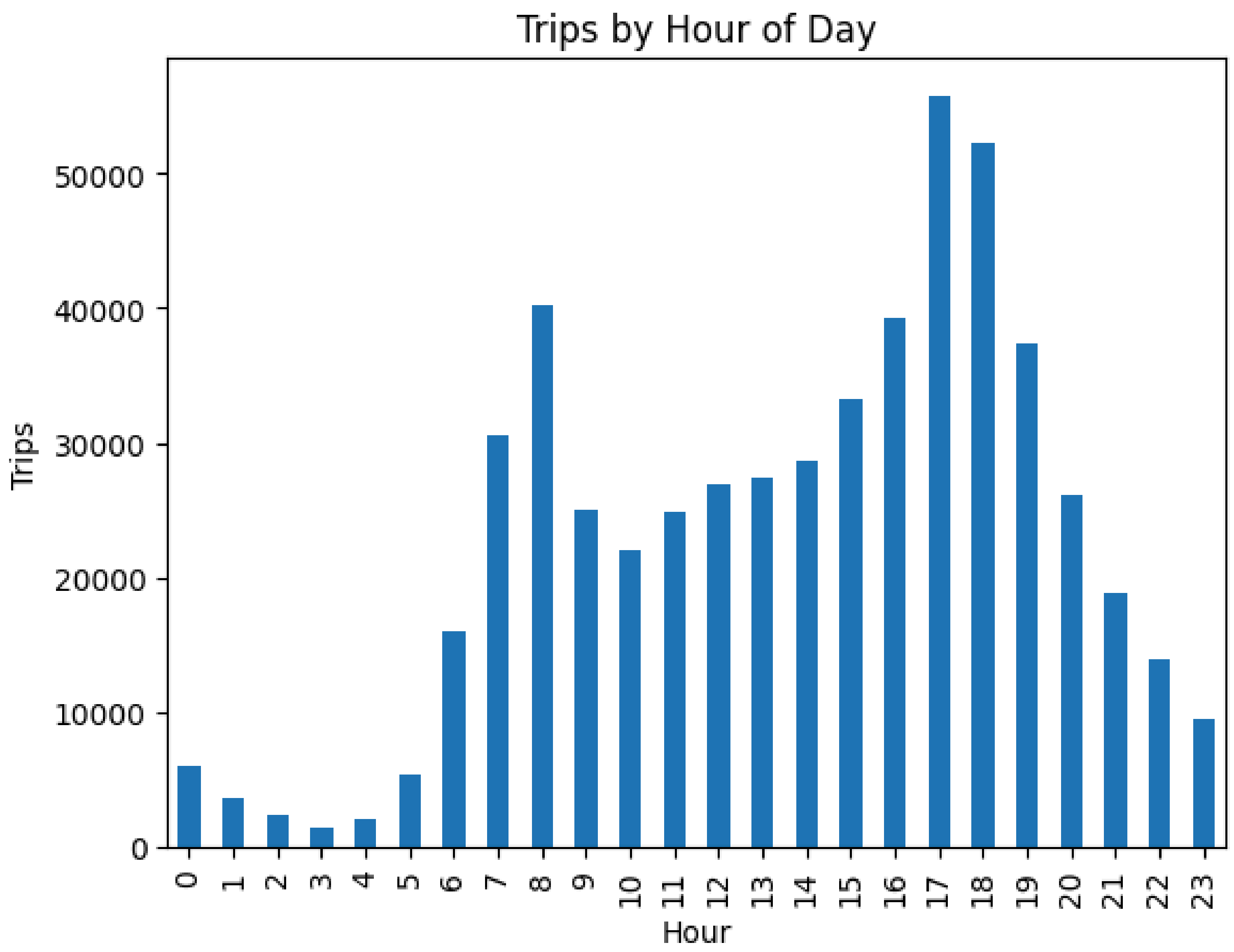
Dataset:Link

```
Data columns (total 13 columns):
 #  Column              Non-Null Count   Dtype
---------------------------------------------------------------
 0  ride_id             548789 non-null  object
 1  rideable_type       548789 non-null  object
 2  started_at          548789 non-null  datetime64[ns]
 3  ended_at            548789 non-null  datetime64[ns]
 4  start_station_name  548788 non-null  object
 5  start_station_id    548789 non-null  object
 6  end_station_name    547308 non-null  object
 7  end_station_id      548789 non-null  object
 8  start_lat           548789 non-null  float64
 9  start_lng           548789 non-null  float64
 10 end_lat             547979 non-null  float64
 11 end_lng             547979 non-null  float64
 12 member_casual       548789 non-null  object
```

EDA

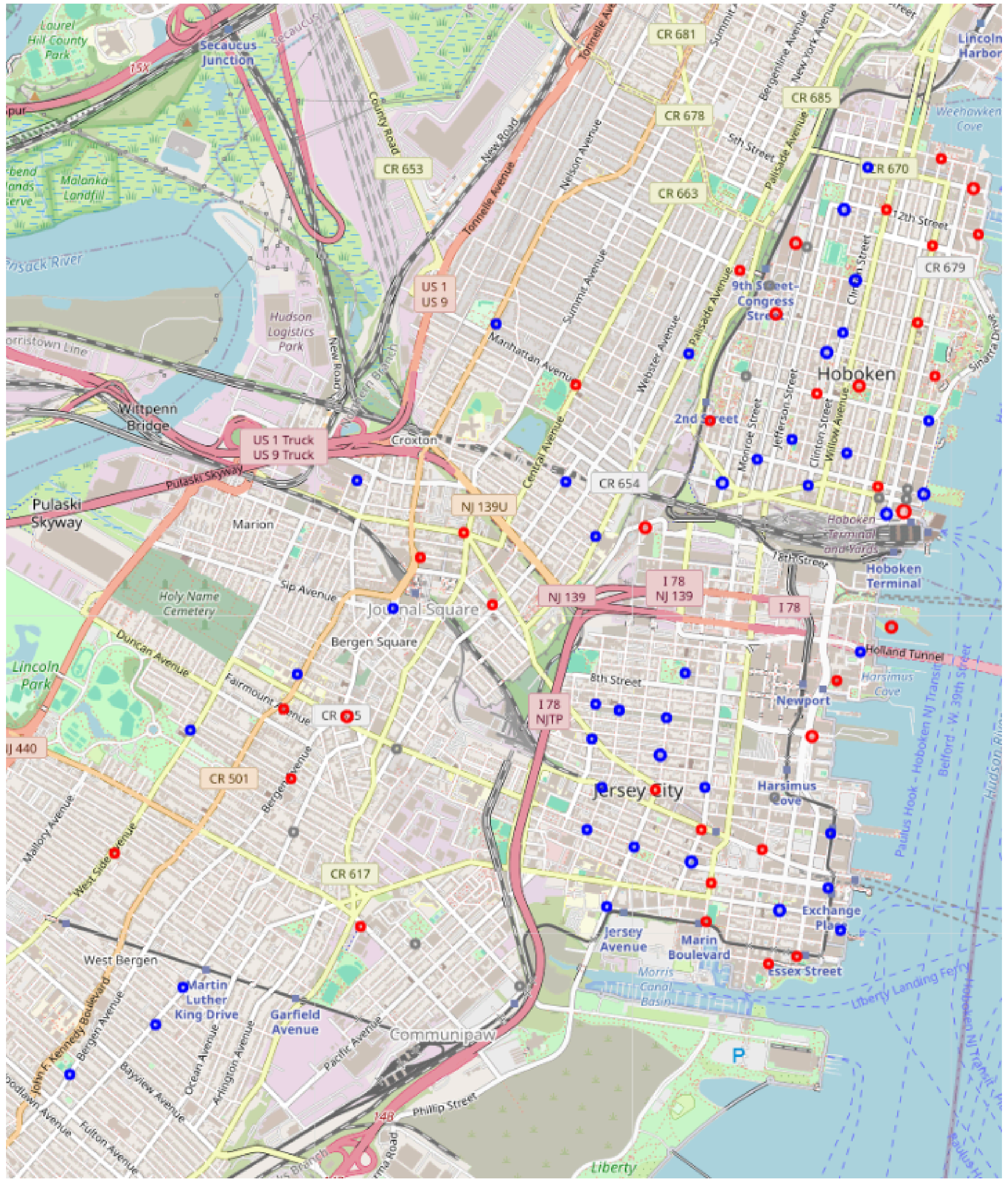


90% <30 minutes ride



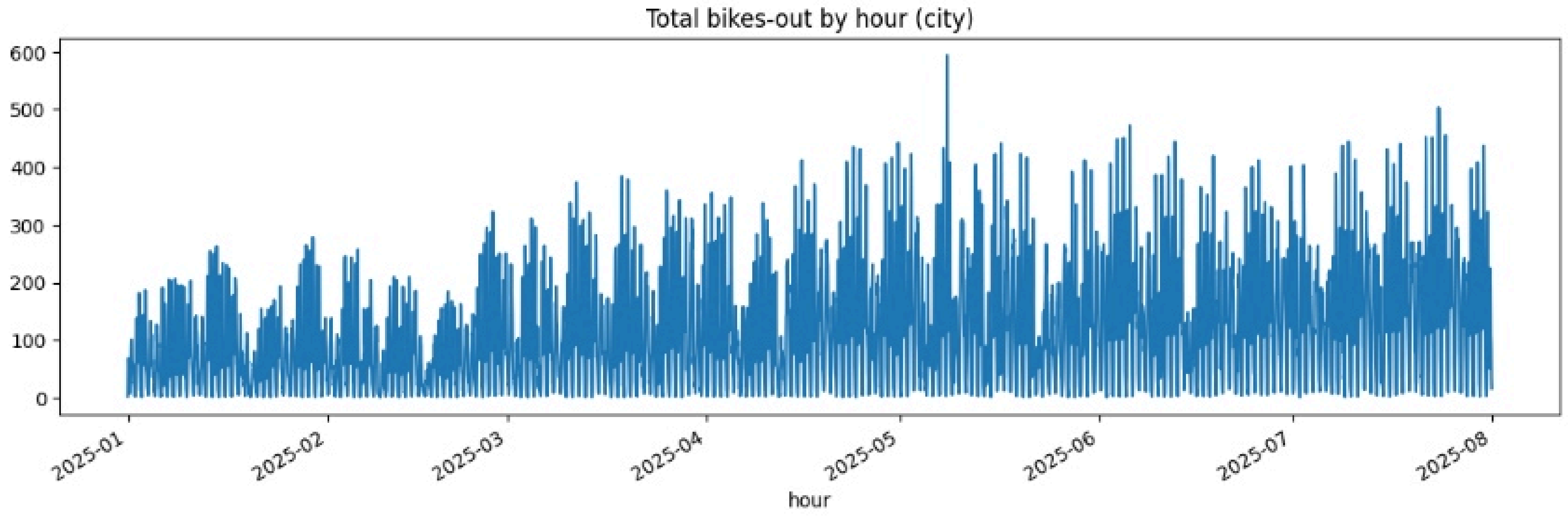
8am-7pm major commute

EDA



Journal Square and Hoboken, New Jersey

blue → surplus regions
red → demanding regions



Frequent marathons from may

EDA

Vehicle Type	Example Name	Capacity (bikes)	Operational Cost per Trip (₹)
Small	Mini Van	10	400
Medium	Light Truck	25	900
Large	Box Truck/Bulk Van	50	1700

Vehicles constraint

03

Methodology

Which methods to utilize for optimal working of system

Why MILP ?

MILP solvers use branch-and-bound or cutting-plane algorithms to search the entire solution space and guarantee globally optimal solutions (or precise bounds when relaxed).

MILP was selected for this project due to its flexibility in capturing all real-world constraints, vehicle types, batching, and operational fairness objectives in a unified and interpretable framework.

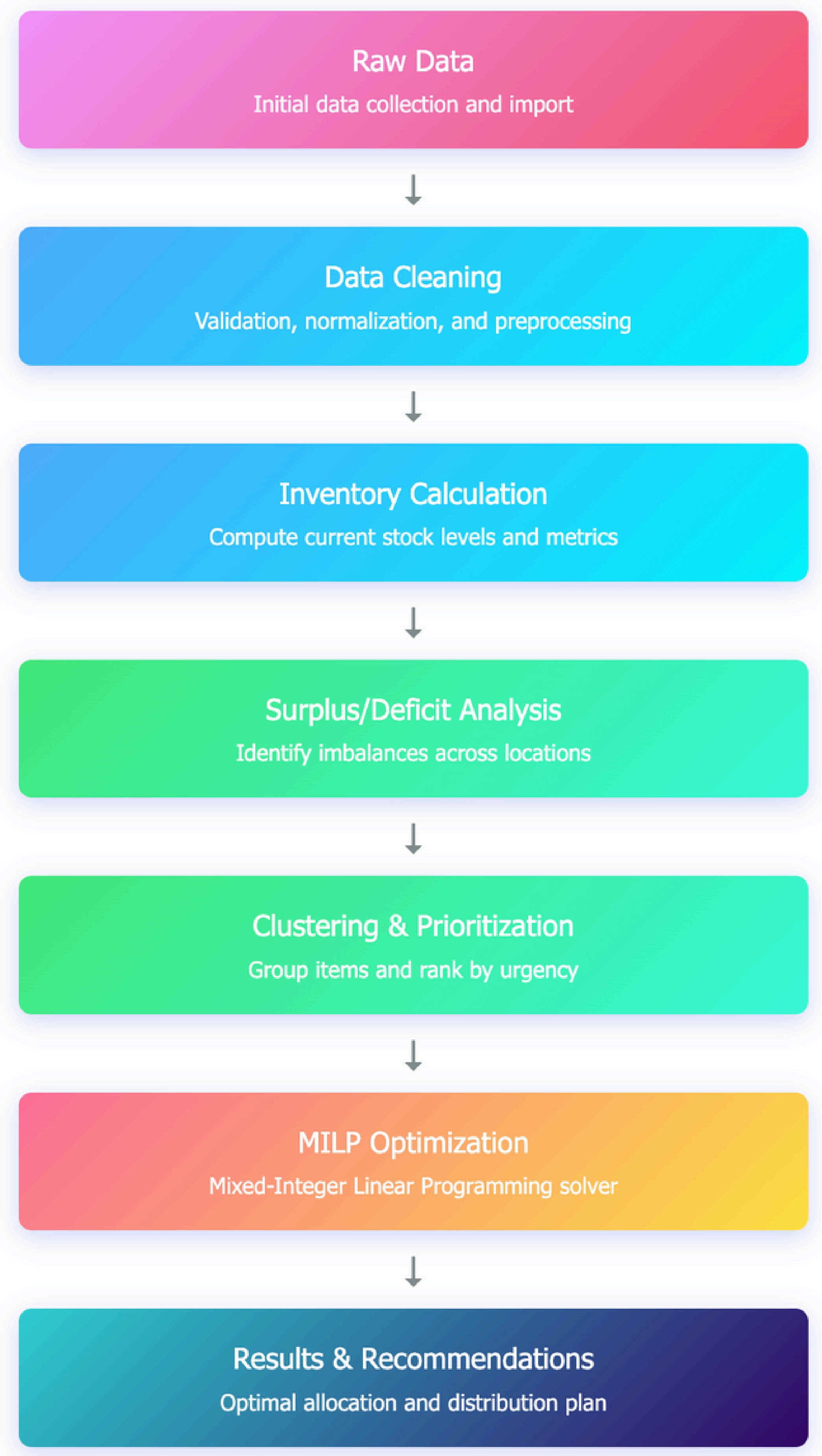
Overall Pipeline

Data Preparation: Aggregate the Citi Bike trip data by station/hour to estimate net bike inflows and outflows across 7 months.

For each deficit station: demand di= deficiti, shortage yi, served si=di−yi, service level SLi=si/di.

Gini: $G = \frac{2\sum_{i=1}^{n} i \cdot SL_{(i)}}{n \sum SL} - \frac{n+1}{n}$,

where SL(i) are sorted ascending. Fairness score = 1−G, clipped to . Higher means more even service across deficit stations.



Inventory and Demand Calculation: For each station and planning window, compute surplus and deficit using formula:

Δi=xi−si, where xi is the required bikes and si is current inventory.

Thank You / DEMO